REINFORCE Algorithm

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1 REINFORCE (Monte Carlo Policy Gradient)

1.1 Goal

Develop a policy-gradient learning algorithm that can improve a parametrized stochastic policy $\pi(a|s,\theta)$ using stochastic gradient ascent.

We seek an update rule for θ based on sample estimates whose expectation is proportional to the true gradient $\nabla J(\theta)$.

1.2 Starting Point: Policy Gradient Theorem

The Policy Gradient Theorem provides:

$$\nabla J(\theta) \propto \mathbb{E}_\pi \left[q^\pi(S_t, a) \nabla \pi(a|S_t, \theta) \right]$$

This expresses the gradient as an expectation under the current policy π.

Expanded form:

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a q^\pi(s,a) \nabla \pi(a|s,\theta)$$

where:

- $\mu(s)$ is the on-policy state distribution.
- $q^\pi(s,a)$ is the action-value function under π.

Key Observation: Sampling from π naturally samples states according to $\mu(s)$.

1.3 First Idea: All-Actions Estimation

One could update θ considering all actions:

$$\theta_{t+1} = \theta_t + \alpha \sum_a \hat{q}(S_t, a, w) \nabla \pi(a|S_t, \theta)$$

where $\hat{q}$ estimates q^π.

Name: all-actions method. **Problem:** requires evaluating all actions; computationally expensive.

1.4 Focusing on Sampled Actions (REINFORCE Path)

Starting from:

$$\nabla J(\theta) \propto \mathbb{E}_\pi \left[\sum_a q^\pi(S_t, a) \nabla \pi(a|S_t, \theta) \right]$$

Apply importance sampling:

$$\nabla J(\theta) \propto \mathbb{E}_\pi \left[\sum_a \frac{\pi(a|S_t,\theta)}{\pi(a|S_t,\theta)} q^\pi(S_t,a) \nabla \pi(a|S_t,\theta) \right]$$

Thus:

$$\nabla J(\theta) \propto \mathbb{E}_\pi \left[q^\pi(S_t,A_t) \frac{\nabla \pi(A_t|S_t,\theta)}{\pi(A_t|S_t,\theta)} \right]$$

where $A_t \sim \pi(\cdot|S_t,\theta)$.

1.5 Final Key Step: Replace $q^\pi(S_t,A_t)$ with G_t

By definition:

$$\mathbb{E}[G_t|S_t,A_t] = q^\pi(S_t,A_t)$$

Thus:

$$\nabla J(\theta) \propto \mathbb{E}_\pi \left[G_t \frac{\nabla \pi(A_t|S_t,\theta)}{\pi(A_t|S_t,\theta)} \right]$$

1.6 Final REINFORCE Update Rule

The stochastic gradient ascent update:

$$\theta_{t+1} = \theta_t + \alpha G_t \nabla \log \pi(A_t|S_t,\theta)$$

where:

$$\nabla \log \pi(A_t|S_t,\theta) = \frac{\nabla \pi(A_t|S_t,\theta)}{\pi(A_t|S_t,\theta)}$$

This term is called the eligibility vector.

1.7 Interpretation

- The eligibility vector points in the direction to increase $\pi(A_t|S_t,\theta)$.
- Scaling by G_t promotes actions yielding higher returns.
- Normalization by $\pi(A_t|S_t,\theta)$ prevents domination by frequently sampled actions.

1.8 Characteristics of REINFORCE

- Monte Carlo method: needs the full return G_t.
- Suitable for episodic tasks.
- Provides unbiased but high-variance estimates.

1.9 Adding Baselines for Variance Reduction

Introduce a baseline $b(S_t)$:

$$\theta_{t+1} = \theta_t + \alpha(G_t - b(S_t))\nabla \log \pi(A_t|S_t, \theta)$$

Properties:

- Expected gradient remains unbiased.
- Baseline reduces variance.

Common baselines:

- Constant baseline.
- State-value function $V^\pi(S_t)$.
- Learned critic model.

1.10 The REINFORCE Algorithm

Input: differentiable policy $\pi(a|s, \theta)$, step size $\alpha > 0$.
Algorithm:

1. Initialize θ.
2. Loop over episodes:
   (a) Generate an episode: $S_0, A_0, R_1, S_1, A_1, R_2, \ldots, S_T$.
   (b) For each step t:
   - Compute return:

$$G_t = \sum_{k=t+1}^{T} \gamma^{k-t-1} R_k$$

   - Update:

$$\theta \leftarrow \theta + \alpha G_t \nabla \log \pi(A_t|S_t, \theta)$$

1.11 Important Notes

- Pseudocode often includes a discount factor γ.
- Using $\nabla \log \pi(a|s, \theta)$ simplifies calculations.
- The eligibility vector captures how the policy is sensitive to θ.

2 REINFORCE with Baseline

2.1 Goal

Extend the REINFORCE algorithm to incorporate a **baseline** into the policy gradient update, aiming to **reduce variance** while preserving the expected direction of the update.

2.2 Starting Point: Policy Gradient Theorem with Baseline

The Policy Gradient Theorem can be generalized as:

$$\nabla J(\theta) \propto \sum_s \mu(s) \sum_a (q^\pi(s,a) - b(s)) \nabla \pi(a|s,\theta)$$

where:

- $b(s)$ is an arbitrary baseline function, independent of action a,
- $\mu(s)$ is the on-policy state distribution.

Why Subtracting $b(s)$ Works

Because:

$$\sum_a b(s) \nabla \pi(a|s,\theta) = b(s) \nabla \sum_a \pi(a|s,\theta) = b(s) \nabla 1 = 0$$

Thus, subtracting a baseline does **not bias** the gradient estimate.

2.3 Update Rule: REINFORCE with Baseline

Following similar derivations as standard REINFORCE, the new update rule is:

$$\theta_{t+1} = \theta_t + \alpha (G_t - b(S_t)) \nabla \log \pi(A_t|S_t,\theta_t)$$

where:

- G_t is the total return from time t,
- $b(S_t)$ is the baseline value at state S_t.

2.4 Effect of the Baseline

- Helps differentiate between high-value and lower-value actions.
- In high-value states, prevents unnecessarily large updates.
- In low-value states, allows appropriately scaled updates.

This mirrors the use of a baseline in the gradient bandit algorithms discussed earlier.

2.5 Choice of Baseline

A **natural choice** for the baseline is an estimate of the state value:

$$\hat{v}(S_t, w)$$

where $w \in \mathbb{R}^d$ is a parameter vector learned during training.

2.6 Full Algorithm: REINFORCE with Learned Baseline

- **Input**: Differentiable policy $\pi(a|s, \theta)$ and value function approximation $\hat{v}(s, w)$.
- **Step sizes**: $\alpha_\theta > 0$, $\alpha_w > 0$.

Algorithm:

1. Initialize θ and w (e.g., both to zero).
2. Loop over episodes:
   (a) Generate an episode $(S_0, A_0, R_1, S_1, \ldots, S_T)$ using π.
   (b) For each time step t:
   - Compute return:

$$G_t = \sum_{k=t+1}^{T} \gamma^{k-t-1} R_k$$

   - Update value parameters:

$$w \leftarrow w + \alpha_w (G_t - \hat{v}(S_t, w)) \nabla \hat{v}(S_t, w)$$

   - Update policy parameters:

$$\theta \leftarrow \theta + \alpha_\theta (G_t - \hat{v}(S_t, w)) \nabla \log \pi(A_t|S_t, \theta)$$

2.7 Notes on Step Sizes

- α_w (value learning step size): relatively easy to choose; can use rules like $\alpha_w = 0.1/\mathbb{E}[\|\nabla \hat{v}(S_t, w)\|^2]$.
- α_θ (policy learning step size): harder to set; depends on reward magnitude and policy parameterization.

2.8 Final Remarks

- The baseline does not bias the policy gradient.
- It significantly reduces variance, leading to faster and more stable learning.
- Learning a baseline with a value function approximation is effective and practical.